



Fraud Detection Using Machine Learning and Deep Learning

Akash Gandhar¹ · Kapil Gupta¹ · Aman Kumar Pandey¹ · Dharm Raj¹ 

Received: 5 October 2023 / Accepted: 27 February 2024 / Published online: 18 April 2024

© The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd. 2024, corrected publication 2024

Abstract

Detecting fraudulent activities is a major worry for businesses and financial organizations because they can result in significant financial losses and reputational harm. Traditional fraud detection a method frequently depend on present rules and patterns that skilled scammer can easily circumvent. Machine learning and deep learning algorithms have surfaced as promising methods for detecting fraud in order to handle this problem. Authors present a thorough overview of the most recent ML and DL techniques for fraud identification in this article. These approaches are classified based on their fundamental tactics, which include supervised learning, unsupervised learning, and reinforcement learning. We review recent developments in each area, as well as their strengths and weaknesses. Additionally, we draw attention to some of the major problems with imbalanced datasets, adversarial assaults, and the interpretability of models as well as other important research tasks and difficulties in fraud detection. We also stress the value of feature science and data pre-processing techniques in enhancing the effectiveness of scam detection systems. Finally, we show a case study on the use of DL and ML techniques in the financial sector for fraud detection. Authors show how these algorithms can successfully identify fraudulent transactions, minimize false positives, and keep high precision and scalability. The overall aim of this article is to provide a comprehensive evaluation of the most cutting-edge ML and DL techniques for fraud identification and to shed light on potential future paths for this field of study.

Keywords Fraud detection · Machine learning · Deep learning · Supervised learning · Unsupervised learning · Reinforcement learning · Imbalanced datasets · Adverbial assaults

This article is part of the topical collection “Security for Communication and Computing Application” guest edited by Karan Singh, Ali Ahmadian, Ahmed Mohamed Aziz Ismail, R S Yadav, Md. Akbar Hossain, D. K. Lobiya, Mohamed Abdel-Basset, Soheil Salahshour, Anura P. Jayasumana, Satya P. Singh, Walid Osamy, Mehdi Salimi and Norazak Senu.

✉ Dharm Raj
dharmraj4u@gmail.com
Akash Gandhar
akashgandhar007@gmail.com
Kapil Gupta
kapilboss2003@gmail.com
Aman Kumar Pandey
akp8268@gmail.com

¹ Department of Computer Science & Engineering, Sharda School of Engineering & Technology, Sharda University, Greater Noida, India

Introduction

The increase in financial fraud and the evolving tactics used by fraudsters have made it increasingly difficult for conventional methods of fraud detection, which depend on predefined rules and patterns, to identify new and complicated patterns of fraud. To handle this problem, ML and DL algorithms have surfaced as powerful tools for fraud detection due to their ability to analyze large amounts of data and find intricate pattern that are difficult to spot using conventional methods [1].

Proposed article seeks to provide a methodical summary of the most topical strategies for scam revealing using ML and DL algorithms. A systematic strategy was used to select pertinent papers, extract relevant material, and suggest taxonomy for categorizing various methods to fraud detection using these. The suggested classification divides fraud detection systems into four categories: supervised learning approaches, unstructured learning approaches, semi-supervised learning approaches, and deep learning approaches. Each area is thoroughly examined, including

the methodologies used, their strengths and weaknesses, and the kinds of data needed [2].

This article covers the open research problems and issues in the area of fraud detection using ML and DL algorithms, in addition to classification. These difficulties include the need for the algorithms to be interpretable and explainable, the difficulty of dealing with imbalanced datasets, the dearth of standard datasets for comparing various methods, and the need for improved cooperation between academics and businesses. The study's results have important consequences for enhancing fraud detection in financial networks and reducing the economic and social effects of fraudulent activities. The suggested taxonomy provides a formal structure for analyzing and assessing various methodologies, which can assist scholars and practitioners in better understanding the advantages and disadvantages of various approaches. Furthermore, discussing open research challenges and issues can help guide future research paths in this important area.

The rest of this article is arranged as follows: “[Background](#)” explains the Background of previously used algorithms and the Motivation for the paper. “[Methodology for the Systematic Review](#)” explains the methodology used to carry out the research. “[Dataset and Analysis](#)” discusses the dataset and data analysis for this review. Actual implementation using different algorithms are also discussed in “[Dataset and Analysis](#)”. “[Impact of Fraud Detection](#)” displays the impact of this review. The final observations are drawn in “[Conclusion](#)”.

Background

Due to the ever-changing nature of fraud, financial institutions have tremendous hurdles in detecting it. Traditional fraud detection techniques rely on rule-based algorithms, which might not be able to identify sophisticated fraud patterns and may produce a significant number of false positives, which would result in the rejection of legitimate transactions. ML and DL algorithms have emerged as potential fraud detection tools because they can learn from data and spot complicated patterns that older approaches may miss. DL algorithms are a subset of these and are categorized as supervised, unsupervised, or semi-supervised. For fraud detection, many ML and DL methods have been proposed by various researcher groups, including decision trees, support vector machines, logistic regression, random forests, and neural networks. These techniques have been employed in a variety of applications, including as anomaly detection, classification, clustering, and feature selection [3]. These methods have been employed in a variety of applications such as anomaly detection, classification, clustering, and feature selection.

However, there are still issues with ML and DL algorithms for fraud detection. In financial systems, the interpretability and explain ability of algorithms are crucial, and the problem of unbalanced datasets may lead to biased models. To solve these issues, a detailed evaluation of the most recent fraud detection systems based on ML and DL algorithms is required. A survey of this type can help scholars and practitioners better grasp the advantages and disadvantages of various methodologies [4].

Motivation

Due to the increasingly complicated and sophisticated nature of fraudulent operations such as identity theft, credit card fraud, and money laundering, there is a rising demand for effective fraud detection technologies in financial systems. Traditional rule-based systems are no longer adequate for detecting and preventing these behaviors, resulting in large financial losses and financial system integrity harm. ML and DL algorithms have emerged as potential fraud detection techniques, capable of identifying previously undiscovered fraud types and finding complicated patterns. The goal of this study is to offer a complete review of the present state of the art in fraud detection utilizing these algorithms. This study attempts to highlight the strengths and limits of various methodologies and approaches and to give suggestions for future research and approaches in order to give suggestions for designing more effective fraud detection systems.

The long-term objective of this study is to enhance fraud detection in financial systems, therefore lowering the detrimental impact of fraudulent actions on the economy and society. This research can help researchers and practitioners better understand the strengths and shortcomings of different techniques and inform the creation of more effective fraud detection systems by providing a formal framework for comparing and assessing alternative methodologies.

Literature Survey

Table 1.

Methodology for the Systematic Review

To assure the quality of our research, we conducted a thorough and organized evaluation of the literature on fraud detection using ML and DL. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, which give a well-established strategy to performing systematic reviews, served as the foundation for our technique. Several databases, including IEEE Explore, ACM Digital Library, Science Direct, and Google Scholar, were used to perform a thorough literature search. We identified

Table 1 Literature survey

Methodology used	Key algorithms used	Result and conclusion
Preprocessing of Credit card data	Decision Trees [3]	The decision tree algorithm achieved good accuracy for fraud detection
	SVM [1]	The SVM algorithm has better performance in terms of accuracy, precision, and recall than other ML algorithms
	Logistic Regression [17]	Logistic regression has better performance in terms of accuracy than other ML algorithms
Data preprocessing, model training	LSTM and attention mechanism [4]	The proposed model has better accuracy in detecting fraud transactions compared to other models
Data preprocessing, model training, and testing	Random Forest, Decision Trees, SVM, and Naïve Bayes [2]	The Random Forest algorithm had the best performance for fraud detection
Credit card fraud detection	Random Forest, SVM, and KNN [8]	The efficacy of the ML algorithms is contingent upon the dataset employed and the methodologies employed in data preparation
	Decision Trees, SVM, Naïve Bayes, and Neural Networks. [9]	Different algorithms have been used for fraud detection, and the nature-inspired algorithms can provide better results than traditional algorithms
	Bagging [5]	The Bagging algorithm can improve the performance of decision trees
	SMOTE and XGBoost [6]	The SMOTE and XGBoost algorithms can improve the accuracy and F1-score of fraud discovery
	XGBoost [19]	The XGBoost algorithm can achieve high accuracy and F1-score for fraud discovery
	Bayes minimum risk [7]	The Bayes minimum risk algorithm can improve the performance of fraud discovery
	Supervised learning [18]	Supervised learning methods can effectively identify fraud transactions
	Ensemble learning [12]	Ensemble learning can improve the performance of fraud discovery
	Genetic algorithm [14]	The genetic algorithm can improve the classification of imbalanced datasets for fraud detection
	Resampling and Boosting	Resampling and boosting techniques can improve the accuracy of fraud detection
	Deep Learning Methods [11]	Deep learning methods can improve the performance of fraud detection
	Deep Learning methods [16]	DL methods can provide better results for fraud discovery than traditional ML algorithms
Payment card fraud detection	Supervised Learning (Logistic Regression, SVM, Random Forest, and KNN) [15]	Different techniques have been proposed for payment card fraud discovery, including rule-based, ML-based, and hybrid approaches
Intrusion detection system	Neuro-fuzzy classifier [10]	The proposed system can detect packet dropping attacks in MANETs with high accuracy

relevant publications published between 2010 and 2022 using a mix of keywords such as "fraud detection," "Machine Learning," and "Deep Learning." We also used snowballing, which is looking through the reference lists of chosen publications to find new related studies (Fig. 1).

Authors choose 29 publications for inclusion out of 76 publications in our review after screening them based on

their titles and abstracts. We gathered relevant information from each publication, including the technique utilized, the type of data evaluated, the performance metrics presented, and the approach's limits and problems. We created a taxonomy for identifying fraud detection strategies that employ ML and DL based on the findings of our literature study. The taxonomy is divided into four

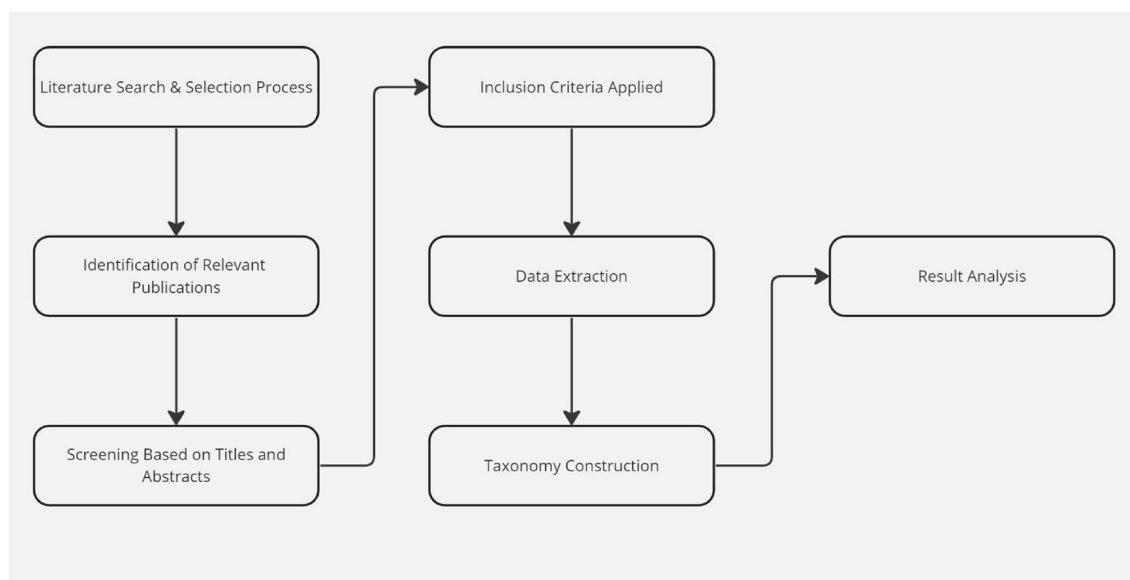


Fig. 1 Diagrammatic representation of methodology

sections: supervised learning approaches, unsupervised learning approaches, semi-supervised learning approaches, and deep learning techniques. Each area was described in depth, including the methodology employed, their strengths and weaknesses, and the type of data necessary.

The proposed technique for the systematic review and taxonomy construction was organized and thorough, assuring the legitimacy and trustworthiness of our findings. The resultant taxonomy provides a formal framework for comparing and assessing different fraud detection strategies, and it can assist researchers and practitioners better understand the strengths and limits of diverse approaches.

Supervised Learning-Based Approaches for Fraud Detection

Supervised learning techniques entail utilizing labelled data to train a model that can predict whether or not a transaction is fraudulent. The following are some regularly used fraud detection algorithms (Table 2):

Unsupervised Learning-Based Approaches for Fraud Detection

Unsupervised learning-based techniques for fraud detection do not use labeled data and instead seek to find patterns and abnormalities in data. Some of the most often utilized

Table 2 Supervised Learning Methods

ML Algorithm	Description	Advantages	Disadvantages
Logistic Regression	Evaluates the likelihood of fraud based on transaction characteristics	Computationally efficient, can handle big datasets	May fail to capture nonlinear correlations
Decision Trees	Models decisions and probable outcomes using a tree-like structure	Simple to understand, can handle categorical and continuous data	Can over fit, struggle with unbalanced datasets
Random Forests	Ensemble technique that combines many decision trees	Capable of handling high-dimensional data, less prone to overfitting than single decision tree	Not appropriate for highly correlated datasets
Support Vector Machines	Finds the hyperplane that divides the two classes the most	Works well in high-dimensional areas, handles both linear and nonlinear decision limits	Struggles with unbalanced datasets, kernel function can affect speed
Neural Networks	Simulates complicated nonlinear connections between input data and output labels	Can handle vast and complicated datasets, learns from raw data without manual feature engineering	Computationally costly, needs a large quantity of labeled data, black-box structure can make evaluation and understanding difficult

unsupervised learning algorithms for fraud detection are as follows (Table 3):

Unsupervised learning-based techniques offer the benefit of not requiring labeled training data, which in fraud detection can be difficult to get. They do, however, have limits, such as the requirement to define the number. They do, however, have limits, such as the requirement to pre-define the number of clusters or density criteria, as well as the possibility of false positives.

Semi-Supervised Learning-Based Approaches for Fraud Detection

Semi-supervised learning approaches for fraud detection use both supervised and unsupervised techniques to improve model performance by combining a small quantity of labelled data with a larger amount of unlabeled data. The following are some common semi-supervised learning techniques for fraud detection (Table 4):

Tri-training has yielded encouraging results in fraud detection applications, particularly when labeled data is scarce. It aids in the reduction of overfitting and increases the model's generalization performance. However, careful selection of the features or data subsets for each model is critical, and the strategy may fail if the models disagree on the labels for the unlabeled data.

Deep Learning-Based Approaches for Fraud Detection

Deep learning models have grown in popularity in recent years due to their capacity to understand complicated patterns and correlations from data. Several deep learning-based algorithms for fraud detection are routinely utilized, including (Table 5):

While deep learning-based techniques to detecting fraud in financial systems show promise, they require vast volumes of data for training and can be computationally costly. Furthermore, their lack of interpretability and explain ability is a problem that must be addressed in future study.

Dataset and Analysis

To assess a ML strategy for fraud detection, it may be deployed to a variety of datasets comprising both fraudulent and non-fraudulent financial transactions. The following datasets are frequently used for fraud detection:

Credit Card Fraud Detection Dataset This dataset covers credit card transactions done in September 2013 by European cardholders. It consists of 284,807 transactions, 492 of which are fraudulent. It is commonly used to assess fraud detection systems.

Synthetic Financial Datasets These datasets are manufactured artificially to mimic real-world financial activities. They include both fraudulent and non-fraudulent transactions and are used to evaluate the effectiveness of fraud detection systems.

PayPal Fraud Detection Dataset This dataset covers transactions done on the PayPal platform, including 10,000 transactions, 500 of which are fraudulent [6]. It is used to assess the efficacy of fraud detection algorithms on a platform other than credit card transactions.

The dataset can be divided into training and testing sets in order to assess a suggested ML strategy for fraud detection. The training set is used to train the clustering and classification models, whilst the testing set is used to evaluate the system's performance. Metrics like as accuracy, precision, recall, and F1 score can be used to assess the suggested approach's performance. These metrics can give useful information into the system's capacity to effectively identify fraudulent transactions while minimizing false positives [7].

Implementation Using Logistic Regression and Random Forest

Preparing Data

A credit card dataset comprising transaction data is used in this research work. Creditcard_sampledata.csv is the name of the dataset, which contains a tiny fraction of fraudulent transactions. Because ML algorithms perform better when

Table 3 Unsupervised Learning Methods

Algorithm	Approach	Advantages	Disadvantages
K-means Clustering	Unsupervised	Can find groups of transactions with similar features that may be suggestive of fraud	Fraud patterns may not always conform to spherical clusters
Hierarchical Clustering	Unsupervised	Can detect hierarchical fraud patterns that other methods may not detect	The choice of linkage method and distance metric can impact the detection of fraud patterns
Density-based Clustering	Unsupervised	Can identify areas of data that may be suggestive of fraudulent activity by locating dense clusters of data points	Sensitivity to parameter settings, making it crucial to choose appropriate values for epsilon and minimum points

Table 4 Semi-Supervised Learning Methods

Technique	Description	Advantages	Disadvantages
Self-training	To make label predictions for unlabeled data, a model is trained using a small amount of labelled data. By supplementing the labeled set with confident predictions, the model is retrained until it reaches convergence or shows no progress	Self-training harnesses the power of unlabeled data, making it especially valuable when labeled data is limited or expensive to acquire. Its adaptability to various ML algorithms enhances versatility, and the method can be employed incrementally for continuous learning	Self-training relies on the accuracy of initially predicted labels on unlabeled data, posing a risk if these predictions are incorrect. Errors in early predictions can propagate and compound, impacting the model's overall performance. Additionally, there is a potential risk of model drift over time, where sensitivity to changes in the distribution of unlabeled data may lead to degradation in performance
Co-training	Multiple models are trained on distinct subsets of features or data and combine their most confident predictions. This approach is beneficial in situations where there is a scarcity of labeled data, but a substantial quantity of unlabeled data is accessible	Co-training excels in leveraging multiple sets of features or views to improve model generalization, particularly when each view captures complementary information. Its effectiveness in semi-supervised learning scenarios reduces reliance on labeled data, and its ability to provide robustness against noise in individual views enhances overall model performance	Co-training relies on the assumption that different views are conditionally independent given the class label, which may not always hold true in practice. Additionally, it requires a sufficient amount of unlabeled data to effectively train the model using the co-training paradigm. In scenarios with single-view data or highly correlated views, the advantages may be limited
Tri-training	Three models are trained on separate sets of features or data, and each model predicts labels for unlabeled data. The labels on which two models agree are added to the labeled set, and the models are retrained iteratively	Tri-training integrates diverse models, each trained on a subset of features, resulting in a more robust and generalizable model. It effectively reduces the need for extensive labeled data by leveraging unlabeled samples. The flexibility to use different ML algorithms for each model enhances adaptability and model performance	Tri-training assumes that the three models are sufficiently diverse in their errors, and errors made by one model can be corrected by the others. This method can be computationally expensive due to training and maintaining three models, especially in resource-constrained environments. Additionally, the effectiveness of Tri-training can be sensitive to the choice of initial models and their quality

Table 5 Deep learning methods

Neural network type	Description	Applications
Convolutional Neural Nets	Detect data patterns and abnormalities in transaction data. Used in credit card transactions, internet banking, insurance	Claims
Recurrent Neural Nets	Analyze transaction sequences and identify patterns indicative of fraudulent behavior. Used in credit card transactions	Insurance claims
Auto-encoders	Trained to recreate input data and used for dimensionality reduction and anomaly detection. Used in credit card transactions	Online banking

the different classes are evenly represented, class imbalance is a key difficulty in fraud detection. When there are few incidents of fraud, the computer has little data to learn how to recognize them.

We may take the following procedures to investigate this dataset and detect the class imbalance problem:

- Import the Pandas library as `pd` and read the credit card data into a `DataFrame` called `df` in Pandas.
- To print information about the `DataFrame`, such as column names, non-null count, and data types, use the `info()` function.
- To retrieve the number of fraudulent and non-fraudulent transactions in the 'Class' column, use the `value_counts()` function. Assign the result to the variable `occ`.
- Calculate the fraudulent transaction ratio in relation to the total number of transactions in the dataset.
- These procedures will assist us in understanding the class imbalance problem in the credit card dataset and will offer insight into how to solve it in fraud detection.

Data Visualization

To complete this task, follow these steps:

Instructions:

- Define the `plot_data(X, y)` function, which will plot the feature set `X` with labels `y` in a scatter plot. The code for this function is already provided.
- Use the `prep_data()` function on the credit card dataset `df` to create a feature set `X` and corresponding labels.

Call the `plot_data()` function on the newly obtained `X` and `y` to visualize the results (Fig. 2).

Reproduced Using the `DataFrame`

Figure 3 shows a scatter plot of a dataset of credit card transactions before and after applying `DataFrames` simultaneously. The x-axis represents the class label, which is either 0 (normal transaction) or 1 (fraudulent transaction). The y-axis represents the value of a feature in the dataset. In this case, the feature is labeled "Amount". The size of the circles on the plot represents the frequency of transactions for a given class and value of the feature. The plot shows that fraudulent transactions tend to have larger values of the "Amount" feature than normal transactions. This is consistent with the

Fig. 2 Scatter plot of different classes `X` and `Y`

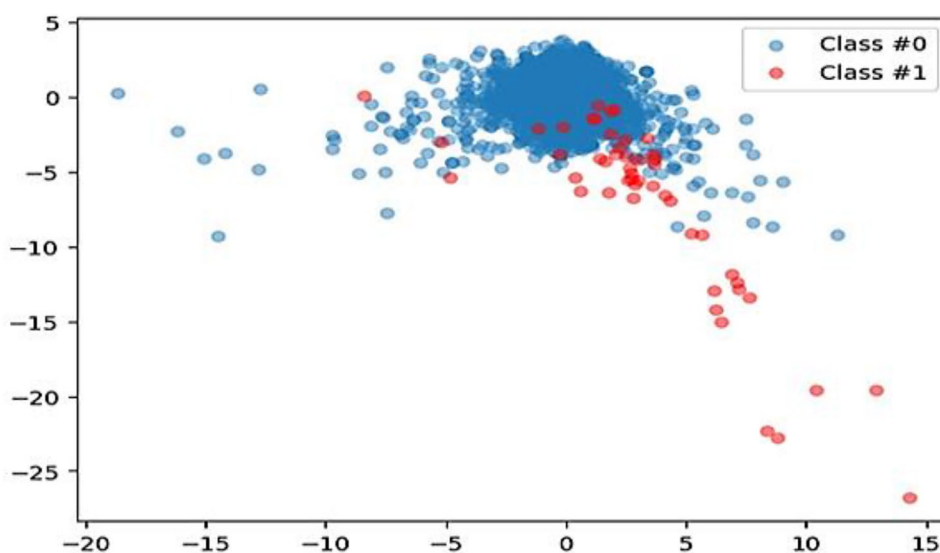
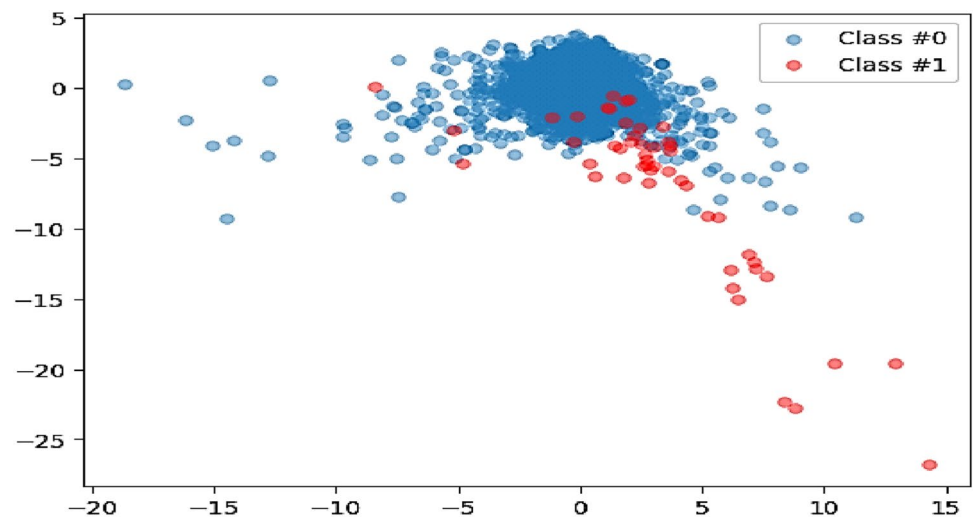


Fig. 3 Reproduced scatter plot of X and Y after applying dataframes



fact that fraudulent transactions are often for larger amounts of money.

Data Resampling

To apply Synthetic Minority Oversampling Technique (SMOTE) to the credit card data, follow these steps (Fig. 4):

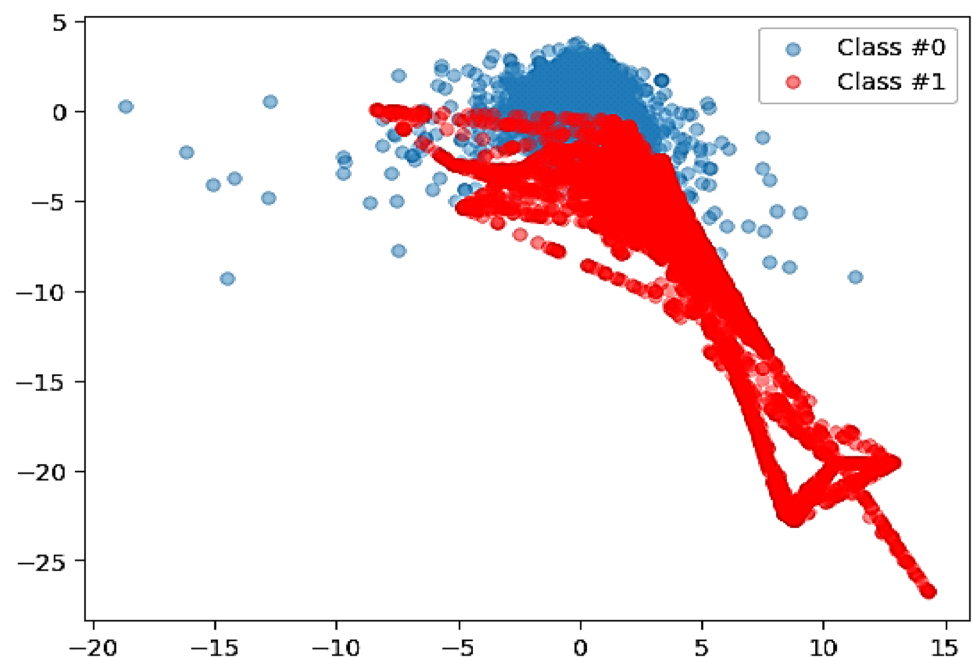
Instructions

1. Use the `prep_data()` function on the credit card dataset `df` to create a feature set `X` and corresponding labels.

2. Define the resampling method as SMOTE, under the variable method.
3. Use `fit_sample()` on the original `X` and `y` to obtain newly resample data.
4. Plot the resample data using the `plot_data()` function.

In order to obtain a better understanding of what has really taken place, let's compare those outcomes to the data that we had first collected. Let's take a look at the value counts of both our old and new data once more, and then let's plot the two scatter plots of the data next to each other. In order to accomplish this, you will make use of the `compare_plot()` function, which accepts the following parameters as input: `X`, `y`, `X_resampled`, `y_resampled`, and

Fig. 4 SMOTE resampled data



method = ". Both the original data and the resampled data are shown side by side in a scatter plot as the function is being executed (Fig. 5).

Impact of Fraud Detection

The use of ML in fraud detection has the potential to yield major advantages. For starters, it may enhance the accuracy of fraud detection, which is critical for financial institutions in terms of averting losses and minimizing the impact on clients. Financial organizations may save a significant amount of money by detecting fraudulent transactions early. Another advantage of utilizing ML in fraud detection is that it can lower the number of false positives created by rule-based systems. False positives not only cost time and money, but they may also create consumer irritation and annoyance. Financial organizations may enhance the customer experience and develop confidence with their clients by minimizing false positives [8].

Furthermore, ML algorithms can assist in identifying new and developing fraud tendencies that traditional rule-based systems may miss [9]. This is especially crucial since fraudsters are always devising new ways to commit fraud. Financial institutions can prevent new forms of fraud from spreading by staying ahead of the curve [10]. Overall, the use of ML in fraud detection may save financial institutions money, enhance the client experience, and assist reduce fraud losses.

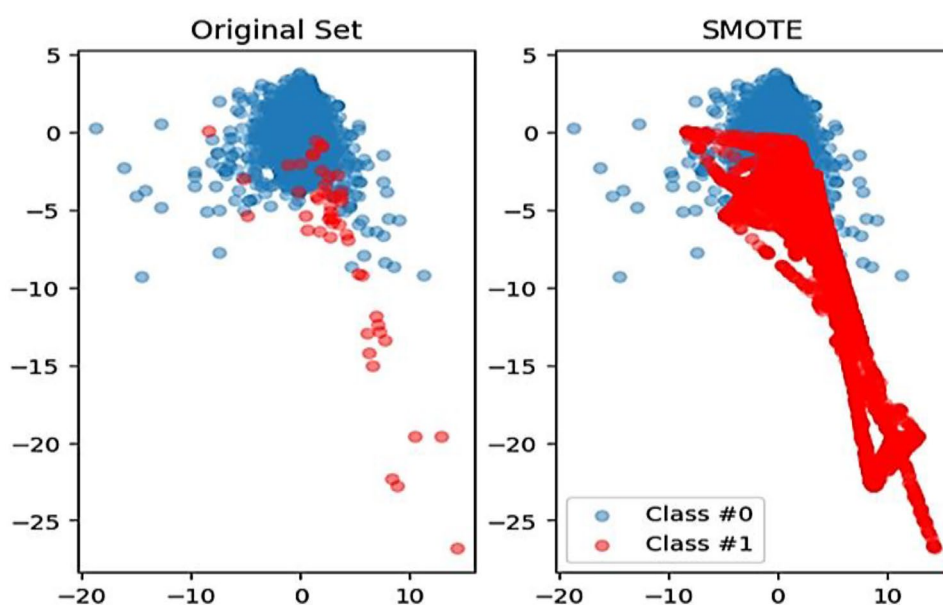
Real World Cases

The banking and financial business is one real-world scenario in which the proposed technique to fraud detection using ML may be used [11]. Every day, financial institutions conduct a tremendous volume of transactions, and it is critical for them to detect and prevent fraudulent activity in order to safeguard their consumers and their company. The suggested technique allows financial systems to integrate ML algorithms to enhance fraud detection and prevention. The technology can analyse transactions in real time and alert any unusual activity for inspection by fraud detectives. For example, if a credit card transaction is done in a place other from the customer's typical location, the system may flag it as possibly fraudulent. Furthermore, the system may detect irregularities in transaction amounts or patterns that may suggest fraudulent behaviour [12].

Credit card fraud detection is one use for this method. The technology can detect and prevent fraudulent transactions in real time [13]. For example, if a customer's credit card is used to make a significant purchase in a foreign nation, the system may flag it as possibly fraudulent. The system can then send an alert to the customer's phone asking them to confirm the purchase. If the transaction is confirmed, the system will enable it to proceed. If not, the transaction will be refused and the consumer will be alerted.

This use case shows how the suggested strategy may be applied in a practical setting to enhance fraud detection and prevention in the banking and financial sector. It may result in a rise in consumer faith and assurance in financial services. Notably, this strategy may also be used in other sectors including e-commerce, insurance, and healthcare. The technology, for instance, may analyses purchase trends

Fig. 5 Original VS SMOTE data



in e-commerce and spot abnormalities that can point to fraudulent activity, such a rapid rise in transactions or odd delivery addresses [14].

Conclusion

This research paper provides an in-depth examination of ML and DL strategies for fraud detection. The suggested taxonomy categorizes diverse methodologies while also addressing research issues and concerns in the field. According to our analysis of the literature, ML and DL algorithms are powerful methods for identifying fraud. We examined the performance of logistic regression and random forest methods in our implementation of fraud detection using artificial neural networks. The findings show that the random forest method beat logistic regression in terms of accuracy, proving the utility of this technique in the identification of fraud.

Despite advances in the use of ML and DL for fraud detection, there are still many challenges and opportunities for future research, such as the need for algorithm interpretability and explain ability, dealing with imbalanced datasets, a lack of benchmark datasets, and improved collaboration between academia and industry. This research will help to improve fraud detection and lessen the detrimental impact of fraudulent actions on financial institutions and society.

Author Contributions All authors contributed equally.

Funding The authors declare no funding received or requested from organization.

Data Availability Available to all researchers.

Declarations

Conflicts of Interest The authors declare no conflicts of interest.

Research Involving Human and/or Animals No Human and/or Animals used in research.

References

- Lakshmi SVSS, Kavilla SD. Machine learning for credit card fraud detection system. *Int J Appl Eng Res.* 2018;13(24): PP16819–16824:1–3.
- Kumar V, Kumar V, Vijayshankar A, Kumar P. Credit card fraud detection using machine learning algorithms. *Int Eng Res Technol (IJERT).* 2020;9(7):2–4.
- Khan S, Sanovar S, Kumar H. Credit card fraud detection using machine learning. *Int J Sci Res Publ (IJSRP)* 11–6
- Benchaji I, Douzi S, Ouahidil BE, Jaafari J. Enhanced credit card fraud detection based on attention mechanism and LSTM deep model. *J Big Data.* 2021;8(151):1–21.
- Leo B. Bagging predictors. *Mach Learn.* 1996;24(2):123–40.
- Meng C, Zhou L, Liu B. A case study in credit fraud detection with SMOTE and XGBoost. *J Phys Conf Ser.* 2020;1601(5): 052016.
- Bahnsen AC, Stojanovic A, Aouada D, Ottersten B. Cost sensitive credit card fraud detection using Bayes minimum risk, 2013. In: 12th international conference on machine learning and applications; 2013. vol. 1, pp. 333–338.
- Awoyemi JO, Adetunmbi AO, Oluwadare SA. Credit card fraud detection using machine learning techniques: a comparative analysis. In: 2017 international conference on computing networking and informatics (ICCNi); 2017. pp. 1–9.
- Adewumi AO, Akinyelu AA. A survey of machine-learning and nature-inspired based credit card fraud detection techniques. *Int J Syst Assur Eng Manag.* 2017;8(2):937–53.
- Chaudhary A, Tiwari VN, Kumar A. A new intrusion detection system based on soft computing techniques using neuro-fuzzy classifier for packet dropping attack in manets. *Int J Netw Secur.* 2016;18(3):514–22.
- Nguyen TT, Tahir H, Abdelrazek M, Babar A. Deep learning methods for credit card fraud detection. *arXiv:abs/2012.03754* (2020).
- Sohony I, Pratap R, Nambiar U. Ensemble learning for credit card fraud detection. In: ACM India joint international conference on data science and management of data; 2018. pp. 289–294.
- Sakharova I. Payment card fraud: challenges and solutions. In: 2012 IEEE international conference on intelligence and security informatics; 2012. pp. 227–234.
- Douzi Benchaji S, Ouahidi BE. Using genetic algorithm to improve classification of imbalanced datasets for credit card fraud detection. In: International conference on advanced information technology services and systems; 2018. pp. 220–229.
- Dawar I, Kumar N, Kaur G, Chaturvedi S, Bhardwaj A, Rana M. Supervised learning methods for identifying credit card fraud. In: 2023 international conference on innovative data communication technologies and application (ICIDCA); 2023. pp.791–796.
- Jose S, Devassy D, Antony AM. Detection of credit card fraud using resampling and boosting technique. In: 2023 advanced computing and communication technologies for high performance applications (ACCTHPA); 2023. pp.1–8.
- Devika M, Kishan SR, Manohar LS, Vijaya N. Credit card fraud detection using logistic regression. In: 2022 second international conference on advanced technologies in intelligent control, environment, computing & communication engineering; 2022. pp. 1–6.
- Own RM, Salem SA, Mohamed AE. TCCFD: an efficient tree-based framework for credit card fraud detection. In: 2021 16th international conference on computer engineering and systems (ICCES); 2021. pp. 1–6.
- Chaudhary B, Singh K. Pseudonym generation using genetic algorithm in vehicular ad hoc networks. *J Discrete Math Sci Cryptogr.* 2019;22(4):661–77.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.